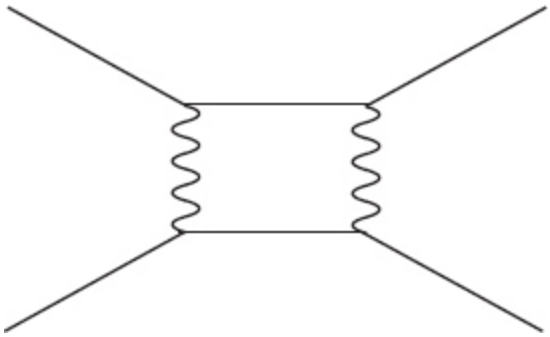


Each line represents a particle with a certain amount of energy. This energy is conserved when lines come together: if one particle comes in and splits into two, for example, the sum of the energies of those two particles must equal that of the initial particle. But how that energy gets split up is completely arbitrary, as long as the sum total is fixed. In fact, due to the wacky logic of virtual particles, the energy of one particle can even be a negative number, such that the other one has more energy than the initial particle did.

This means that when we calculate the process described by a Feynman diagram with an internal closed loop, an arbitrarily large amount of energy can be traveling down any particular line within the loop. Sadly, when we do the calculation for what such diagrams contribute to the final answer, the result can turn out to be infinitely large. That's the origin of the infamous infinities plaguing quantum field theory. Obviously the probability of a certain interaction can be at most 1, so an infinite answer means we've taken a wrong turn somehow.

Feynman and others managed to work out a procedure for dealing with these infinities, now known as *renormalization*. When you have a bunch of quantum fields that interact with one another, you can't simply first treat them separately, and then add in the interactions at the end. The fields are constantly, inevitably affecting one another. Even when we have a small vibration in the electron field, which we might be tempted to identify as a single electron, there are inevitably accompanying vibrations in the electromagnetic field, and indeed in all the other fields that the electron interacts with. It's like playing a piano note in a showroom with many pianos present; the other instruments will begin to gently hum along with the original one, causing a faint echo of whatever notes you are playing. In Feynman-diagram language, this means that even an isolated particle